

A Robust Patch-Based Hierarchical Framework for Image Forgery Detection and Classification

Team Number: 39

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1. Literature Survey & Research Direction

Rubric: Survey Extensiveness & Justification

- **Survey Scope:** Analyzed traditional, CNN-based, and Transformer-based image forgery detection methods.
- **Key Findings:**
 - **Block/Keypoint methods:** Sensitive to geometric transforms; fail in smooth regions.
 - **Global CNN models:** Capture semantic features but miss subtle local forgery artifacts.
 - **Patch-based CNNs:** Better local analysis, but limited to spatial domain only.
 - **Standard ViTs:** Use single-stream spatial tokens—*ignore* frequency and noise forensic cues.
 - **Frequency/Noise methods:** Effective individually, but never fused as *first-class tokens* inside a Transformer.
- **Research Gap:** No existing work fuses spatial, frequency, and noise-residual patch representations as parallel token streams within a Vision Transformer.
- **Direction:** A **Three-Stream Tokenization ViT** with cross-attention fusion for hierarchical forgery detection + classification.

2. Originality of Work

Rubric: Novel Idea vs. Extension

Novelty Statement — Three-Stream Tokenization (Core Contribution)

Novel multi-stream token design for Vision Transformers:

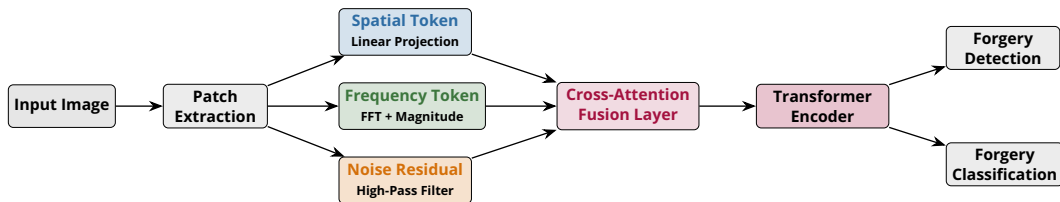
- **Stream 1 — Spatial Token:** Standard patch pixels projected into embedding space (baseline ViT representation).
- **Stream 2 — Frequency Token:** FFT magnitude spectrum of each patch as a separate token embedding. Exposes JPEG blocking, resampling traces, and copy-move boundaries invisible in RGB.
- **Stream 3 — Noise Residual Token:** Gaussian high-pass filtered noise residual (SRM-style) capturing disrupted camera noise patterns at forgery boundaries.

Why This Is Novel

Unlike existing methods that use frequency or noise only as preprocessing, our approach elevates them to *first-class tokens* fused via **cross-attention** alongside spatial tokens before the Transformer encoder.

3. Three-Stream Tokenization Architecture

Novel Multi-Domain Patch Tokenization with Cross-Attention Fusion



Each image patch generates three forensically-rich tokens that are fused via cross-attention before global Transformer reasoning.

4. Technical Knowledge (Algorithms/Models)

Rubric: Demonstrated Knowledge of Fundamentals

Proposed Model Architecture:

- **Post-Processing Robustness:** Dataset augmented with JPEG recompression, Gaussian blur, noise injection, and rescaling to ensure the model generalizes across real-world post-processing conditions.
- **Backbone:** Vision Transformer (ViT) with **Three-Stream Tokenization**
- **Stream Details:**
 - **Spatial:** Flatten patch $\rightarrow$ Linear projection $\rightarrow z_s \in \mathbb{R}^d$
 - **Frequency:** FFT(patch) $\rightarrow |\mathcal{F}|$ magnitude $\rightarrow$ Linear projection $\rightarrow z_f \in \mathbb{R}^d$
 - **Noise:** Gaussian HP filter $\rightarrow$ Residual $\rightarrow$ Linear projection $\rightarrow z_n \in \mathbb{R}^d$
- **Fusion:** Cross-attention across $[z_s, z_f, z_n] \rightarrow$ fused token z^*
- **Hierarchical Strategy:**
 - Stage 1: Binary Forgery Detection (Authentic vs. Tampered)
 - Stage 2: Forgery-Type Classification (Copy-Move, Splicing, Inpainting)
- **Aggregation:** Patch-level predictions $\rightarrow$ image-level decision
- **Explainability:** Attention map visualization + Grad-CAM for forensic interpretation

5. Three Streams & Cross-Attention Fusion

Forensic Motivation + Lightweight Fusion Mechanism

Spatial Stream

- Raw pixel information
- Semantic content & visible artifacts
- Edge & color inconsistencies

Frequency Stream

- FFT magnitude spectrum
- JPEG blocking & resampling traces
- Double compression & copy-move boundaries

Noise Stream

- High-pass noise residual (SRM-style)
- Camera sensor noise patterns (PRNU)
- Disrupted noise at forgery boundaries

Forgery artifacts invisible in one domain are often prominent in another.

Cross-Attention Fusion:

- Stack per-patch: $T_i = [z_s, z_f, z_n] \in \mathbb{R}^{3 \times d}$
- Fuse: $z_i^* = \text{CrossAttn}(T_i) \in \mathbb{R}^d$
- Each stream attends to the other two
- Fused tokens + positional embeddings $\rightarrow$ Transformer

Properties:

- **Lightweight:** Single cross-attention layer
- **Adaptive:** Learns most informative domain per patch
- **End-to-end:** Jointly trainable with Transformer

6. Dataset Details

Training, Evaluation and Forgery Examples

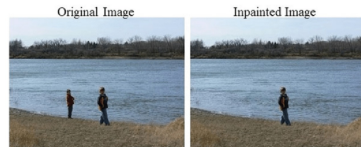
Dataset	Year	# Images	Forgery Types	Purpose
CASIA v2.0	2013	~12K	Copy-Move, Splicing	Primary training dataset
IMD2020	2020	~2010	Inpainting, Removal, Mixed	Training for inpainting and modern edits
Columbia	2004	363	Splicing	Cross-dataset evaluation (high-quality images)
COVERAGE	2018	200	Copy-Move	Cross-dataset evaluation (realistic scenarios)



Copy-Move Forgery



Splicing Forgery



Inpainting Forgery

7. Comparison with Existing Work (1/2)

Research Gap Analysis

Paper / Method		Primary Task	Supervision Type		Major Limitation
Block-based Methods (Fridrich, 2003)		Copy-Move Detection	Image-level	block matching	Sensitive to rotation, scaling, and compression
Frequency-aware CNN		Forgery Detection	Image-level labels		Frequency used only as pre-processing, no joint learning
Standard Transformer	Vision	Image Classification / Detection	Image-level labels		Uses only spatial tokens; ignores forensic noise and frequency cues
ManTra-Net (2019)		Manipulation	Local-ization	Pixel-level supervision	Feature concatenation only; no unified multi-domain reasoning

7. Comparison with Existing Work (2/2)

Research Gap Analysis

Paper / Method		Primary Task	Supervision Type		Major Limitation
MVSS-Net (2022)		Detection + Localization	Pixel-level	supervision	Multi-view multi-scale CNN; no frequency-domain tokens or Transformer reasoning
TruFor (CVPR 2023)		Detection + Localization	Pixel-level	supervision	Two-stream (RGB + noise) Transformer; lacks frequency-domain stream and cross-attention token fusion
Proposed Stream ViT	Three-	Detection + Classification	Type	Image-level labels (patch aggregation)	Addresses limitation: Joint spatial, frequency, and noise reasoning via cross-attention

References

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Thank You